Copenhagen Business School

Home assignment

Predictive Analytics (CDSCO1005E)

Forecasting Electric Vehicle Registrations in Denmark: A Structural Time Series Analysis

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1. Introduction

The adoption of electronic vehicles has become a prominent factor in the transition towards a more sustainable transportation system. Denmark, like many other European countries, has introduced a lot of policy incentives and environmental goals, which aims to accelerate the transition from fossil fuel vehicles to electric alternatives. As a result, the registrations of electric cars has experienced a significant growth over the last decade, which makes it a relevant and timely case for predictive analysis.

The aim of this study is to predict the monthly electric car registrations in Denmark using statistical time series models answering the question "Which statistical time series model provides the most accurate forecasts for monthly electric vehicle registrations in Denmark?" I am to identify the underlying patterns in the data and provide forecasts for the next three years by analysing 170 monthly observations from 2011 to early 2025.

To achieve the objectives, I used and compared widely used univariate time series forecasting models: the AutoRegressive Integrated Moving Average (ARIMA) model and the Exponential Smoothing State Space Model (ETS). Before modeling, I performed several pre-processing steps, including transformations for stationarity, structural break detection, seasonal-trend decomposition. I tested and evaluated multiple model configurations based on diagnostics such as ACF, PACF, and seasonal differentiation requirements. The structural break analysis using the supF test helped to account for possible regime shifts, allowing estimation on both pre-break and post-break segments as well as the full series. The performance of the models was evaluated based on residual diagnostics, prediction performance and accuracy metrics. The best specification

was selected for each model, followed by a final comparison to determine the most suitable forecasting approach and a final forecast was generated.

2. Literature Review

Recent studies have explored diverse methods for forecasting sales of electric vehicles. Kemala et al. (2024) did a comprehensive overview of forecasting methods used in 2011-2023, highlighting the wide variety of approaches. They concluded that causal methods are used most frequently, statistical time series are also common and machine learning techniques are gaining traction.

Peng et al. (2015) used triple exponential smoothing to forecast EV demand in the US, while Dhankhar et al. (2024) analyzed EV sales in India with ARIMA and Holt's exponential smoothing resulting in contradictory findings. Lu (2024) applied ARIMA and SARIMA models to forecast battery electric vehicle (BEV) sales in the UK, where the SARIMA model better captured seasonal fluctuations, highlighting the importance of seasonality in electric vehicle forecasts. These univariate approaches revealed upward trends (often linear or exponential), reflecting the expansion of the market. Zhang et al. (2017) compared univariate and multivariate time series models to forecast EV sales in China. The univariate model used Singular Spectrum Analysis, while the multivariate model applied a Vector Autoregressive approach, incorporating exogenous variables such as government subsidies, fossil fuel prices, and charging infrastructure availability. The inclusion of external variables significantly improved the accuracy of forecasts, highlighting their importance.

The reviewed studies demonstrate the variety of methods used for forecasting EV sales, from classical statistical models to more complex multivariate approaches. While SARIMA proved to efficient in modeling seasonality, others have shown varying results depending on context. These papers reveal that despite differing approaches, there is a consistent growth in EV adoption and emphasizes the need for tailoring forecasting methods to the specific data characteristics and market conditions.

3. Data Description

The dataset used in this analysis is downloaded from the website of Statistics Denmark, called "BIL51: New registrations of passenger cars by ownership and propellant". It consists of 170 monthly observations of new registrations of passenger cars categorized by ownership and propellant type in Denmark from January 2011 to February 2025.

The raw data was obtained as a CSV file and several preprocessing steps were applied using Python, including an initial cleanup, renaming the main column to "Propellant" and setting it as the index. Missing values were replaced with NaN, and date-like column headers (e.g., '2011M01') converted to datetime objects. Finally, the dataset was reshaped from wide to long format, resulting in three columns: "Propellant," "Date," and "Registrations". The cleaned dataset was exported and loaded into RStudio for further analysis.

4. Methodology

4.1 Model Selection

Observing the characteristics of the time series, see Figure 1., it is clear that the data exhibits a strong long-term upward trend with some noticeable seasonal fluctuations, and increasing variance over time. These features suggest that both ARIMA and ETS models are suitable for forecasting. The ARIMA model effectively handles non-stationary time series by differencing and can model complex patterns using autoregressive and moving average components (Box et al., 2015). While the ETS model, is well-suited for series with trend and seasonal components by explicitly modeling different combinations of error, trend, and seasonality, and is thus particularly well suited for the dynamic and seasonal nature of the registration of electric vehicles in Denmark (Hyndman et al., 2021).

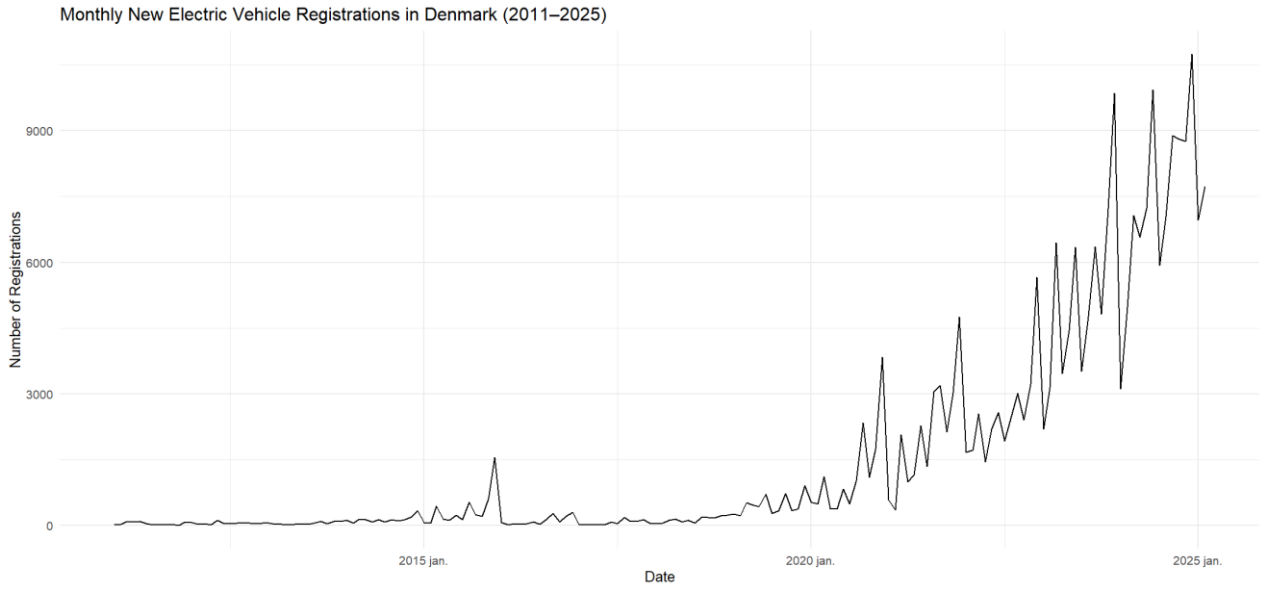


Figure 1: Monthly EV Registrations Time Series

The analysis focuses on univariate forecasting, as the aim is to model and forecast the behaviour of EV registration based on historical values only. This approach allows underlying temporal patterns such as trend, seasonality and autocorrelation structures to guide the forecast without the inclusion of external variables or covariates.

4.2 Data Preparation and Transformation

Before applying forecasting models, the raw data was pre-processed to meet the assumptions required by ARIMA and ETS, aiming to stabilize variance, ensure stationarity, and remove noise and structural patterns that could bias model estimation. For the purpose of this analysis, the dataset was filtered to include only electric vehicles and structured for time series analysis using a monthly index.

Variance Stabilization

The raw time series showed signs of non-constant variance, especially with the rapid increase in later years. To stabilise the variance, a Box-Cox transformation was performed, which is particularly suitable for time

series with strong seasonal patterns. The optimal value of the transformation parameter λ was selected using **Guerrero's method.** This transformation helps to stabilize variance and make the data more normally distributed (Box & Cox, 1964). It takes the form:

$$y(\lambda) = \begin{cases} \frac{y^{\lambda}-1}{\lambda}, & \lambda \neq 0 \\ \log(y), & \lambda = 0 \end{cases}$$

It improved homoscedasticity while preserving the temporal structure of the data.

Seasonal-Trend Decomposition

Seasonal-Trend decomposition using LOESS (STL) was applied to the time series both before and after the Box-Cox transformation to analyse the underlying structure. Decomposition of the original data shows a clear upward trend and annual seasonality, with the seasonal component remaining relatively stable in both shape and amplitude. However, the remainder component shows increasing variance over time, suggesting heteroscedasticity. This pattern confirms that the series is driven by both long-term growth and recurring seasonal cycles, supporting the use of both ETS and seasonal ARIMA models.

After Box-Cox transformation, STL decomposition shows a smoother trend, preserved seasonal structure, and notably more stable residuals. This indicates a successful reduction in heteroscedasticity, producing residuals closer to white noise.

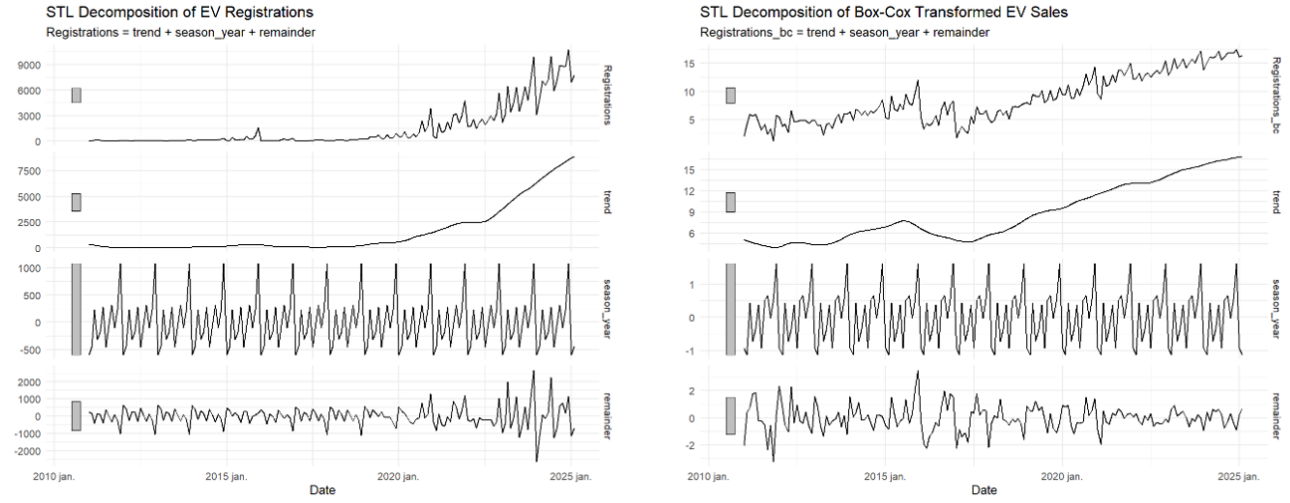


Figure 2:STL decomposition before and after Box-Cox transformation

Structural Break Detection

The detection of structural breaks was performed using the Quandt Likelihood Ratio (QLR) test, also known as the supF test. This test evaluates whether there are significant structural changes in the data at an unknown breakpoint. The analysis was first performed on the non-differenced series and the first and last 15% of observations were excluded. The test returned an extremely low p-value (≈0), indicating strong evidence of a structural break, which was identified at observation 115, corresponding to July 2020. This aligns well with the visible shift in registration growth patterns, likely reflecting external policy shifts or market acceleration. Based on this evidence, I explored two different forecasting strategies:

1. **Structural break modelling**: A level-shift dummy variable was introduced from July 2020 onward, allowing the full dataset (2011–2025) to be retained while accounting for the structural change. This approach assumes that while the level of the series changed, past dynamics still contain predictive power.
2. **Segmented modeling**: The data was split at the detected structural break in July 2020. Forecasting models are trained only on the post-break segment, under the assumption that the break marks a fundamental change in dynamics, making earlier observations less relevant for future predictions.

Trend and Seasonality Removal

Given the long term upward trend, the increasing variance and the inspection of the ACF plot, which shows a slow and gradual decline while the autocorrelations remain significantly positive through many lags, it is clear that the series is not stationary. Differentiation was applied to ensure stationarity.

For the full series, first order differencing was used to eliminate the trend and achieve stationarity. After segmentation at the identified structural break (July 2020), each segment was also assessed independently. The pre-break segment requires only first-order differencing, indicating a trend without seasonal effects. In contrast, the post-break segment shows both trend and seasonality, as detected by the unitroot_nsdiffs() function. Therefore, a seasonal difference with a lag of 12 was applied, followed by a first-order difference to address the underlying trend.

To assess whether the differencing affected the underlying structure, the supF test was reapplied to the differenced full series, which achieved a p-value of 1.0, indicating that no structural break remained. This result reflects the nature of differencing in removing level and trend shifts and further supports the segmentation strategy of post-break modeling.

Stationarity Testing

After the transformations, stationarity was assessed using two formal statistical tests, the Augmented Dickey-Fuller (ADF) Test and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) Test, both before and after differencing. The ADF test is used to determine whether a time series has a unit root, which indicates non-stationarity (Dickey et al., 1979), while the KPSS test is used to test for stationarity around a deterministic trend (Kwiatkowski et al., 1992). Unlike the ADF test, the KPSS takes stationarity as the null hypothesis.

Both tests were applied to the Box-Cox transformed series before differencing. The ADF test was defined with a deterministic trend (type = "trend"), because the original series exhibited a clear upward trajectory. It rejected the null hypothesis of a unit root at 1% significance level, suggesting stationarity. However, the KPSS test contradicted this by rejecting the null of stationarity, indicating the presence of a unit root. These contradicting results imply that the series may follow a stochastic trend rather than a purely deterministic one. This discrepancy is common, since the ADF has low power against trend-stationary alternatives, while KPSS is more sensitive to deviations from trend stationarity, especially in the presence of structural changes. To resolve this ambiguity and ensure full stationarity, differencing was performed. First-order differencing was sufficient to remove the trend from the entire series, which was confirmed by both tests. The pre-break segment only required first-order differencing, the stationarity of which was confirmed by both the ADF and KPSS tests.

The post-break segment exhibited stronger seasonality and required an additional seasonal difference (lag = 12) before achieving stationarity. Again, both statistical tests confirmed the effectiveness of this transformation. Finally, visual inspection of the ACF and PACF plots for the full and segmented series supports the statistical findings, showing no remaining trend or seasonal components, suggesting effective removal of non-stationarity.

4.3 Model Estimation and Evaluation

Two major approaches were considered based on the structural break detected in July 2020: (1) modeling the entire series with a level-shift dummy variable, and (2) modeling only the post-break segment under the assumption of a regime shift.

ARIMA Model Specification

The AutoRegressive Integrated Moving Average (ARIMA) models were fitted using both automated selection and manual diagnostics based on the ACF and PACF plots. Several configurations were tested for each modeling approach, combining various orders of non-seasonal and seasonal autoregressive (AR), differencing (I), and moving average (MA) terms. Model performance was evaluated using the corrected Akaike Information Criterion (AICc) and Bayesian Information Criterion (BIC), as well as residual diagnostics such as the Ljung-Box test to assess white noise behaviour.

Full-Series Model with Structural Break Dummy

To incorporate the structural break identified in July 2020, a level-shift dummy variable (Break) was included in the models, allowing the full dataset (2011–2025) to be used while adjusting for the regime shift. Based on the ACF/PACF diagnostics, an initial **ARIMA(2,1,3)(1,0,1)[12]** model was selected. To validate and compare, 8 different configurations were tested, shown in Table 1. The best-performing model based on the performance criteria proved to be **ARIMA(1,1,2)(1,0,1)[12] with no drift.** Ljung-Box tests confirmed no significant autocorrelation in residuals, indicating an adequate fit.

Model	Drift	AICc	BIC	Ljung-Box p-values
ARIMA4= (2,1,3)(1,0,1)[12]	Yes	554.0	584.0	0.0302
ARIMA1= (1,1,2)(1,0,1)[12]	Yes	555.0	580.0	0.0540
ARIMA5= (1,1,2)(1,0,1)[12]	**No**	**555.0**	**576.0**	**0.0661**
ARIMA3= (2,1,2)(1,0,1)[12]	Yes	556.0	583.0	0.0382
ARIMA2= (1,1,3)(1,0,1)[12]	Yes	557.0	584.0	0.0415
ARIMA6= (1,1,3)(1,0,1)[12]	No	557.0	581.0	0.0519
ARIMA7= (2,1,2)(1,0,1)[12]	No	557.0	581.0	0.0537
ARIMA8= (2,1,3)(1,0,1)[12]	No	558.0	585.0	0.0412

Table 1:Results of different ARIMA specifications with structural break dummy

Post-Break Model (Segmented Approach)

Based on diagnostics, the post-break segment required both a seasonal difference (lag = 12) and first-order differencing. Visual analysis of ACF and PACF plots initially indicated a **ARIMA(2,1,1)(0,1,0)[12]** structure, however ten models were evaluated and compared. The best performing model was **ARIMA(0, 0, 1)(1, 1,**

0)[12] with drift, offering the lowest AICc and strong residual diagnostics. Despite being a simpler model, it fit the truncated data better, with significantly better performance metrics and much lower residual variance. The model effectively captured the strong upward trend and seasonal dynamics, with a significant drift term (≈1.98) and well-behaved residuals. This highlights the model's ability to capture short-term dependencies and seasonality in the post-break period, which significantly improves forecast accuracy.

Model	Drift	AICc	BIC	Ljung-Box p-values
ARIMA1 = (0,0,1)(1,1,0)[12]	**Yes**	**107.0**	**113.0**	**0.395**
ARIMA4 = (1,1,1)(1,1,0)[12]	No	110.0	115.0	0.539
ARIMA7 = (2,1,0)(0,1,0)[12]	No	110.0	115.0	0.840
ARIMA8 = (2,1,0)(1,1,0)[12]	No	110.0	116.0	0.823
ARIMA9 = (2,1,1)(0,1,0)[12]	No	111.0	117.0	0.973
ARIMA2 = (0,1,1)(0,1,0)[12]	No	112.0	115.0	0.607
ARIMA6 = (1,1,2)(0,1,0)[12]	No	113.0	119.0	0.712
ARIMA3 = (1,1,1)(0,1,0)[12]	No	114.0	118.0	0.466
ARIMA5 = (0,1,2)(0,1,0)[12]	No	114.0	118.0	0.341
ARIMA10 = (2,1,2) (0,1,0)[12]	No	114.0	121.0	0.960

Table 2: Results of different ARIMA specifications for post-break model

While the model with a structural break dummy effectively captured the level shift, the post-break model achieved significantly lower error variance and better information criteria, suggesting that modeling the post-break section separately offers a better forecasting strategy in the current context.

ETS Model Specification

The Exponential Smoothing State Space Model (ETS) was applied as an alternative forecasting method. ETS models explicitly capture error, trend, and seasonal components and are particularly well-suited for series with stable seasonality and evolving trend behaviour. Since the ETS framework does not support exogenous regressors, such as the structural break dummy variable, the ETS model was fitted on the post-break segment only, in line with the second modeling strategy. Several ETS configurations were evaluated by varying error, trend and seasonal components:

Model	AICc	BIC	Ljung-Box p-value
ETS(A,N,A)	**163**	**169**	**0.167**
ETS(A,A,A)	172	169	0.0000310
ETS(M,N,M)	182	189	0.0679
ETS(A,A,N)	183	190	0.00000250
ETS(M,A,M)	183	190	0.000675

Table 3: Results of different ETS model specifications

These models were compared based on their corrected AICc and BIC, and residual diagnostics using the Ljung-Box test to assess autocorrelation. The best-fit model selected for the post-break data was an **ETS(A,N,A)**, indicating **additive error**, **no trend**, and **additive seasonality**, which aligns well with the STL decomposition, that showed stable seasonality and no clear linear trend in the post-break period (after differencing). The model's smoothing parameters showed a relatively high value for **alpha (0.702)**, indicating significant

weighting of recent observations in the level component, while **gamma was close to zero**, suggesting stable seasonality. The Ljung-Box test (p=0.167) further confirmed the model adequacy indicating no significant autocorrelation.

Model Comparison

In order to evaluate and compare model performance, each selected forecasting model was applied to a test set to generate out-of-sample forecasts. The full-sample ARIMA model with a structural break dummy (ARIMA(1,1,2)(1,0,1)[12]) was retrained on data from 2011 to 2022 and used to forecast the test period (2023–2025). A post-break ARIMA(0,0,1)(1,1,0)[12] with drift was estimated using data from July 2020 to December 2023, while the ETS(A,N,A) model was fitted to the same post-break period and evaluated on the remaining observations. Forecast accuracy was assessed using standard metrics including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE), along with visual inspections of predicted versus actual values.

Model	ME	RMSE	MAE	MPE	MAPE	MASE	RMSSE
ARIMA5 = (1,1,2)(1,0,1)[12] Break dummy	1,77	2,05	1,88	10,9	11,7	1,13	1,02
ARIMA1 = (0,0,1)(1,1,0)[12] Post-break series	**0,0892**	**0,502**	**0,404**	**0,535**	**2,49**	**0,24**	**0,25**
ETS(A,N,A)	1,31	1,54	1,31	8,07	8,07	0,778	0,763

Table 4:Final model comparison

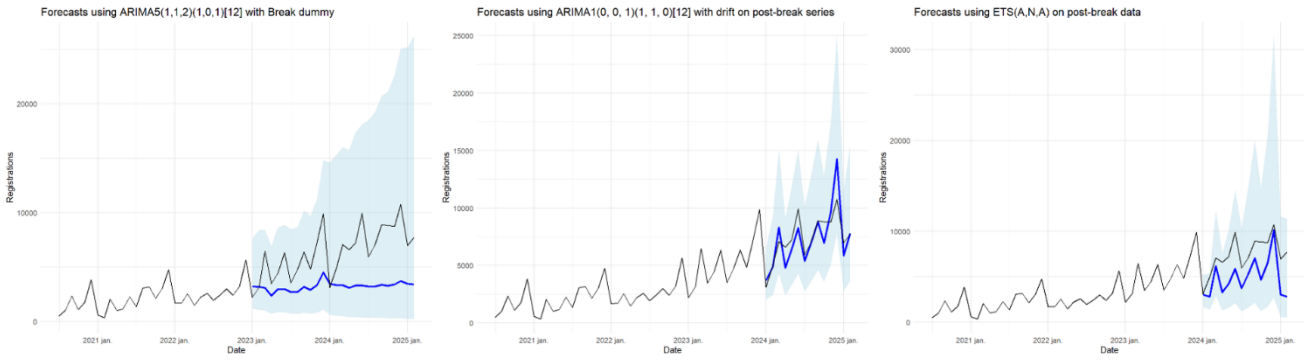


Figure 3:Forecast results on training set with 95% confidence intervals

The full-series ARIMA model with a break dummy showed moderate forecasting performance with a MAPE of 11.7%, MAE of 1.88, and a MASE of 1.13, which is slightly lower than the seasonal naïve benchmark. The model showed a small positive bias (ME = 1.77), suggesting a tendency to overestimate. While the model successfully captured broad seasonal patterns and level shifts, its relatively higher variance may have reduced its precision in later years.

In contrast, the post-break ARIMA model exhibited significantly stronger forecasting performance, achieving a MAPE of 2.49%, MAE of 0.40, and a very low MASE of 0.24, which is well below the seasonal naïve benchmark. Residuals showed no significant autocorrelation, and confidence intervals remained tight across the forecast horizon. This model effectively captured the evolving structure of the series post-2020 and

maintained a stable upward forecast trajectory aligned with market trends. The visual alignment between historical and forecasted values also confirms its effectiveness.

The ETS(A,N,A) model performed competitively, producing a MAPE of 5.4% and MASE of 0.47. The model successfully captured seasonal fluctuations but lacked a drift or trend component, which may have limited its ability to forecast continued growth. Residual diagnostics showed well-behaved errors with no autocorrelation, supporting the validity of model structure.

In summary, while all three models provided valuable forecasts, the segmented ARIMA model applied to the post-break data delivered the most accurate results and most effectively captured the structural dynamics of the electric vehicle registration series. Due to its outstanding predictive accuracy and statistical robustness, this model was selected as the final forecasting model.

5. Forecast Results

After evaluating multiple model configurations and forecasting strategies, the **post-break ARIMA(0,0,1)(1,1,0)[12] with drift** was selected as the final model due to its superior accuracy and residual diagnostics. This model was tailored to capture the dynamics of the EV registration series following the structural shift identified in July 2020. Using this model, a 3-year forecast was generated to project electric vehicle registrations until February 2028, which was performed on the Box-Cox transformed scale to uphold model assumptions and then back-transformed to the original scale using the estimated λ parameter.

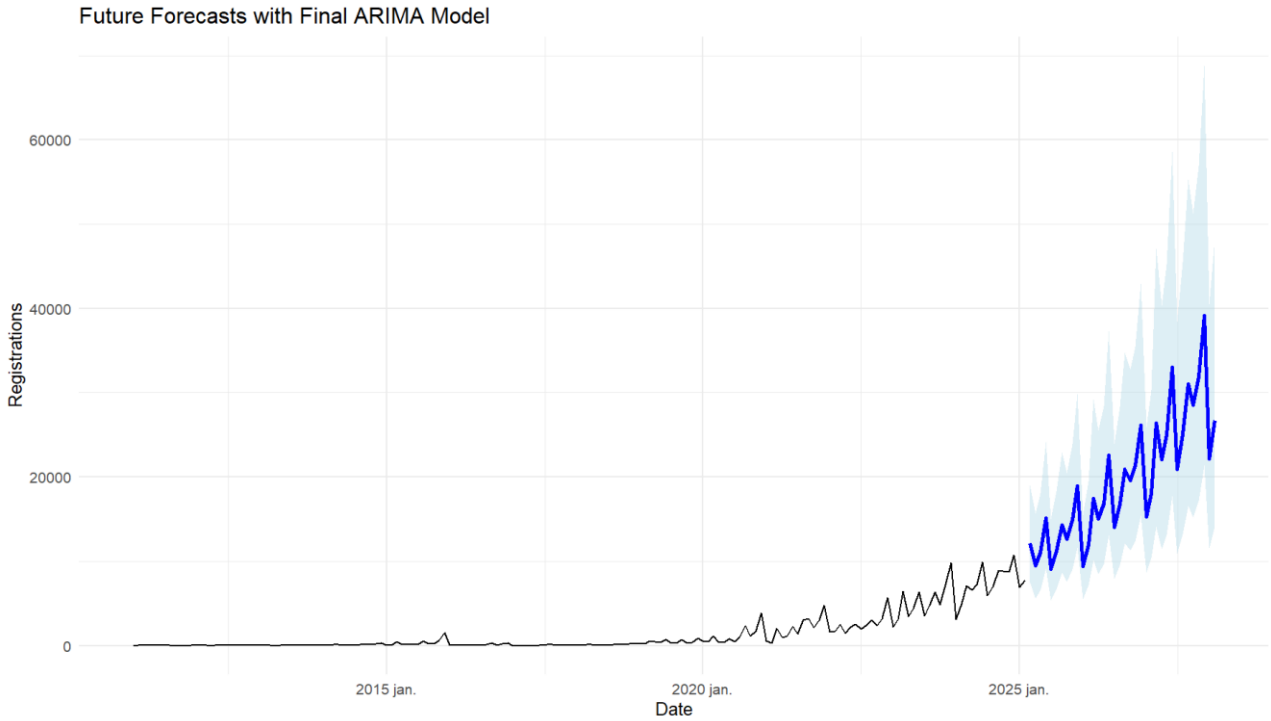


Figure 4:Final Future Forecast

As shown in the plot, the forecast exhibits a clear upward trend with increasing seasonal fluctuations, consistent with the post-2020 growth trajectory and acceleration in EV adoption. Although the 95% confidence intervals

gradually widen over time, reflecting the uncertainty inherent in long-term forecasts, they remain relatively narrow, which indicates strong model stability.

6. Conclusion

The aim of study is to forecast monthly electric vehicle registrations in Denmark using univariate time series models, specifically ARIMA and ETS. The analysis was carried out on a dataset containing 170 monthly observations from 2011-2025 and several pre-processing steps were applied, such as Box-Cox transformation, structural break detection, seasonal-trend decomposition and stationarity testing. A significant structural break has been identified in July 2020, which justified the evaluation of both full-series and segmented modeling strategies. I tested multiple model configurations, including ARIMA models with and without structural break dummies and ETS models with varying error, trend and seasonal components. Model performance was evaluated using standard accuracy measures such as MAE, RMSE, MASE and MAPE, and residual diagnostics were used to validate model assumptions.

The results showed that the segmented ARIMA model applied to the post-break data outperformed both the full-series ARIMA model with level shift dummy and the ETS models. The post-break model achieved the highest forecast accuracy, successfully capturing the new regime's seasonal and structural dynamics. As a result, this model was selected as the final forecasting tool. A final 3-year forecast projected continued growth in EV registrations until early 2028, consistent with the upward momentum observed after the 2020 shift. This final forecast provides a valuable basis for stakeholders seeking to anticipate future electric vehicle demand and infrastructure needs and also underline the importance of identifying and modeling structural changes in time series data—particularly in fast-changing markets such as electric mobility.

In summary, this study highlights the effectiveness of segmented time series modeling in improving forecast accuracy and provides useful insights about EV market trends in Denmark. Future research could explore multivariate approaches incorporating exogenous variables such as policy measures, fuel prices, or infrastructure investments to further enhance predictive ability.

References

Box, G. E. P., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2015). *Time Series Analysis: Forecasting and Control* (5th ed.). Wiley.

Box, G. E. P., & Cox, D. R. (1964). An analysis of transformations. *Journal of the Royal Statistical Society: Series B (Methodological)*, 26(2), 211–252.

Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association, 74*(366), 427–431. https://doi.org/10.2307/2286348

Dhankhar, S., Dhankhar, N., Sandhu, V. et al. (2024). Forecasting Electric Vehicle Sales with ARIMA and Exponential Smoothing Method: The Case of India. *Transp. in Dev. Econ. 10,* 32. *https://doi.org/10.1007/s40890-024-00216-y*

Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice.* OTexts.

Kemala, B. K. L., Surjandari, I., & Puspita, A. N. G. (2024). Forecasting Methods for the Electric Vehicle Ownership: A Literature Review. *Procedia Computer Science, 234, 87-95.* https://doi.org/10.1016/j.procs.2024.02.155

Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., & Shin, Y. (1992). Testing the null hypothesis of stationarity against the alternative of a unit root: How sure are we that economic time series have a unit root? *Journal of Econometrics, 54(1–3), 159–178.* https://doi.org/10.1016/0304-4076(92)90104-Y

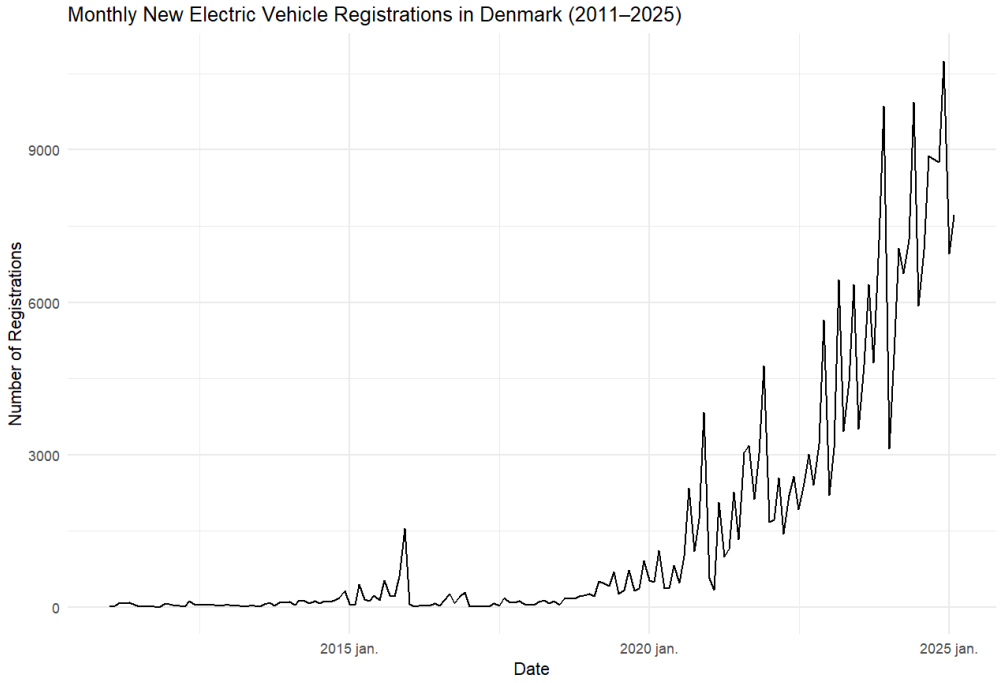
Lu, Yourong. (2024). UK Battery Electric Vehicle Sales Prediction Using ARIMA and SARIMA Models. *Advances in Economics, Management and Political Sciences. 126. 185-190.* 10.54254/2754-1169/2024.18340.

Peng, Zhanglin & Yu, Zhijun & Wang, Hongbo & Yang, Shanlin. (2014). Research on industrialization of electric vehicles with its demand forecast using exponential smoothing method. *Journal of Industrial Engineering and Management. 8.* 365-382. 10.3926/jiem.1287.

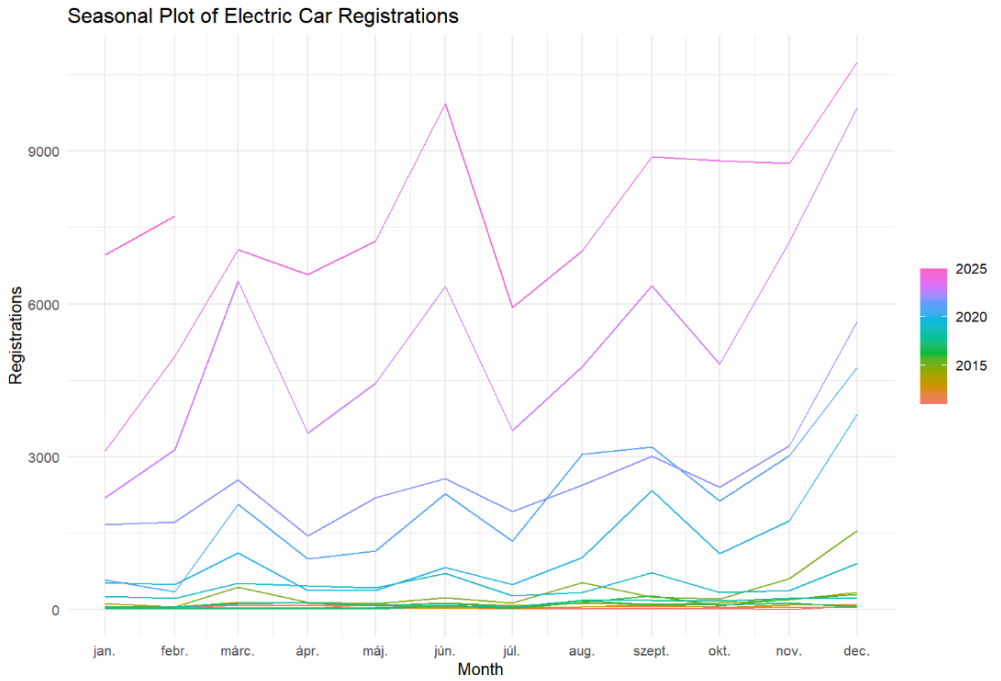
Zhang Y, Zhong M, Geng N, Jiang Y (2017) Forecasting electric vehicles sales with univariate and multivariate time series models: The case of China. *PLoS ONE 12(5): e0176729.* https://doi.org/10.1371/journal.pone.0176729

Appendices

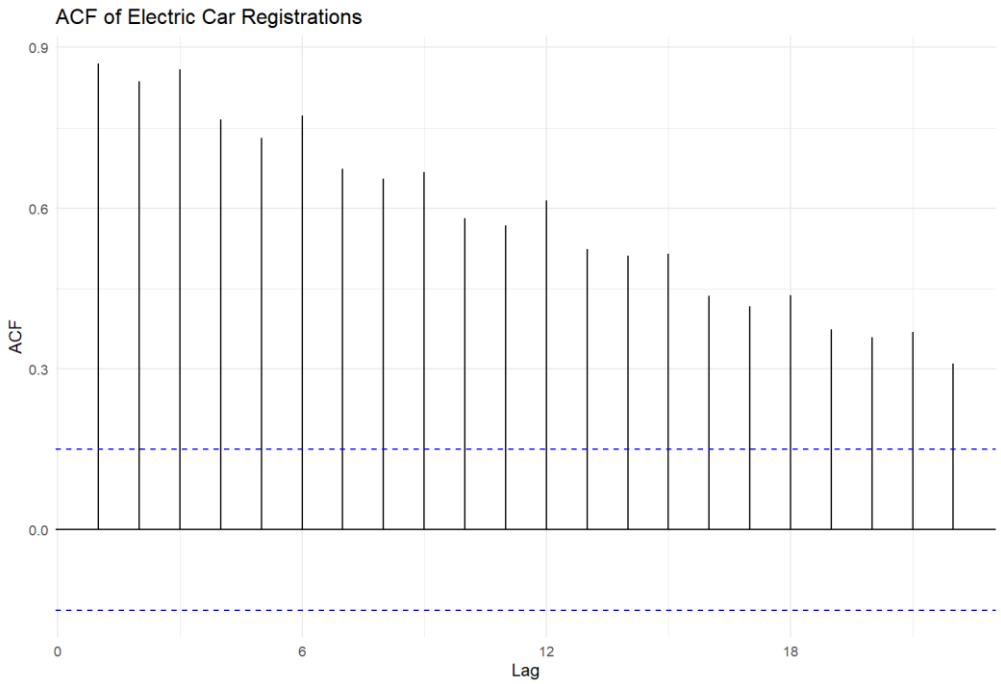
Raw time series plot



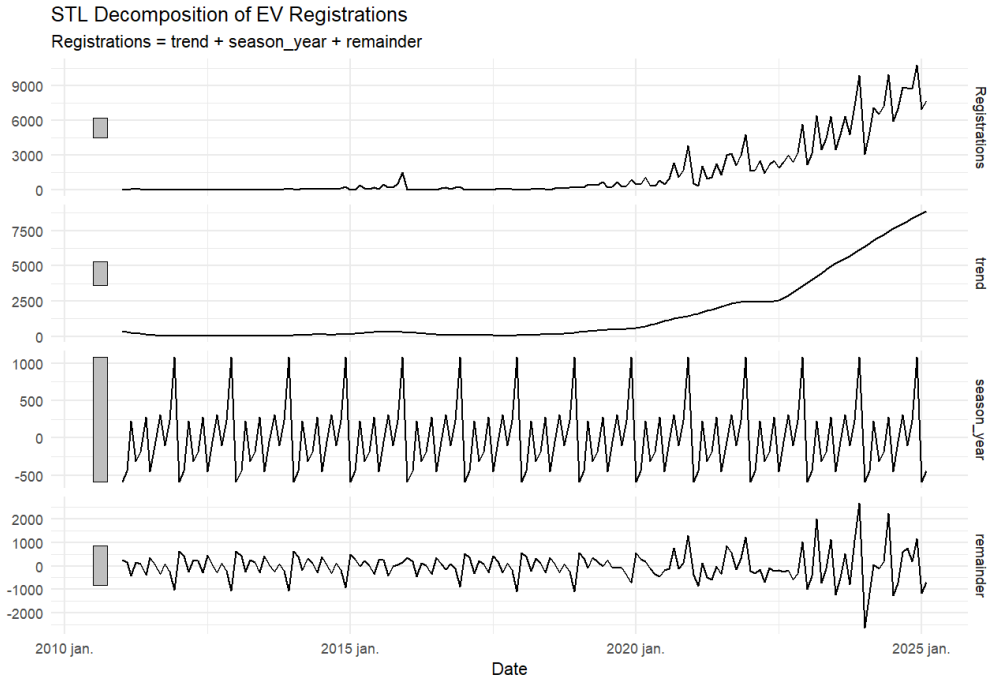
Seasonal plot of raw time series



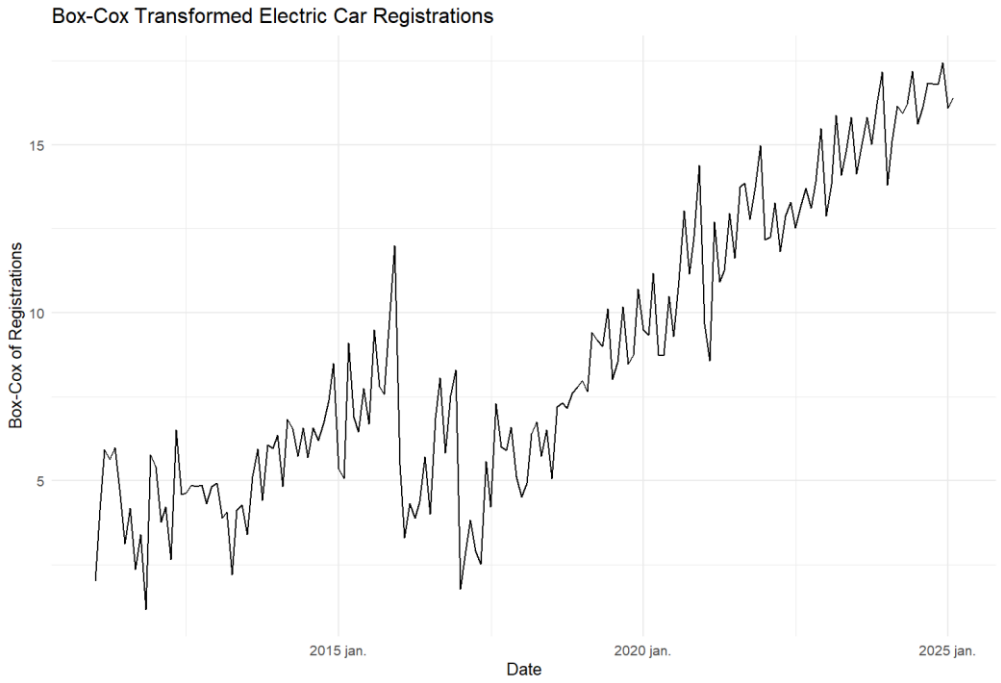
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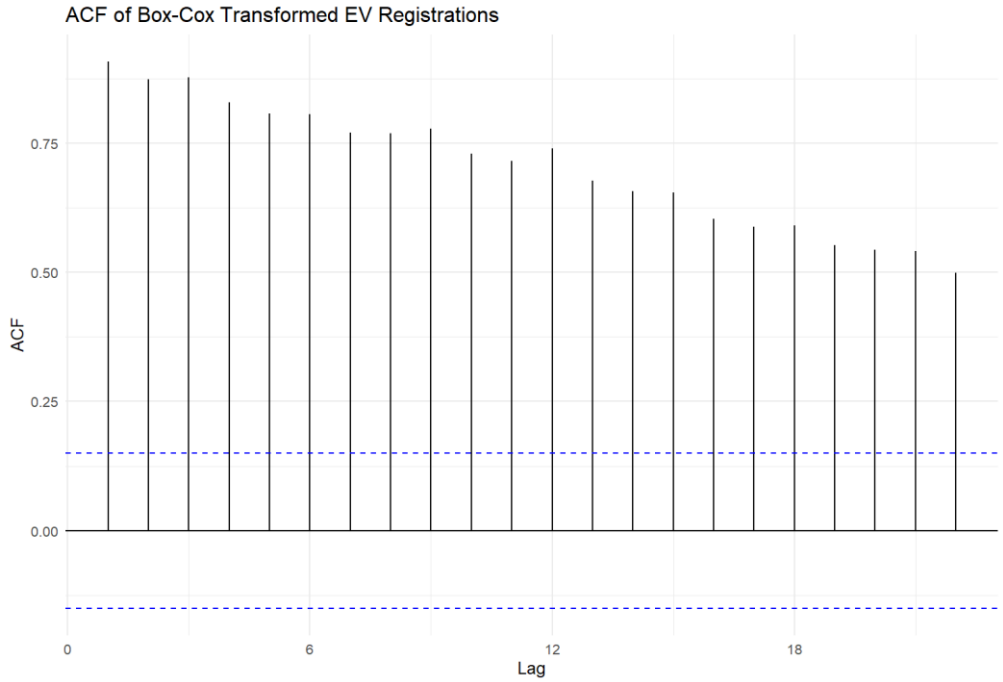
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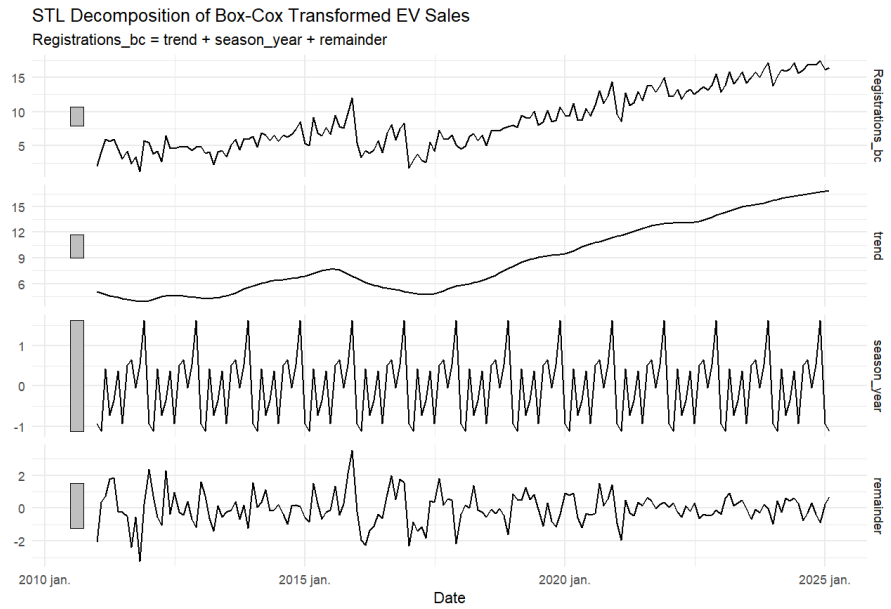
Box-Cox transformed time series



ACF of Box-Cox transformed time series



STL decomposition of Box-Cox transformed time series



Results of KPSS and ADF test applied on the Box-Cox transformed time series

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###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression trend


Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)

Residuals:
    Min      1Q  Median      3Q     Max
-6.2470 -0.8168  0.1000  0.9979  3.6550

Coefficients:
             Estimate Std. Error t value
(Intercept)  0.606892   0.267964   2.265
z.lag.1     -0.276307   0.063958  -4.320
tt           0.022053   0.005482   4.023
z.diff.lag  -0.217701   0.075877  -2.869
            Pr(>|t|)
(Intercept)  0.02483 *
z.lag.1     2.69e-05 ***
tt          8.75e-05 ***
z.diff.lag   0.00466 **
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
  0.1 ' ' 1

Residual standard error: 1.495 on 164 degrees of freedom
Multiple R-squared:  0.2174,    Adjusted R-squared:  0.203
F-statistic: 15.18 on 3 and 164 DF,  p-value: 9.143e-09


Value of test-statistic is: -4.3202 6.5166 9.3774

Critical values for test statistics:
      1pct  5pct 10pct
tau3 -3.99 -3.43 -3.13
phi2  6.22  4.75  4.07
phi3  8.43  6.49  5.47
```

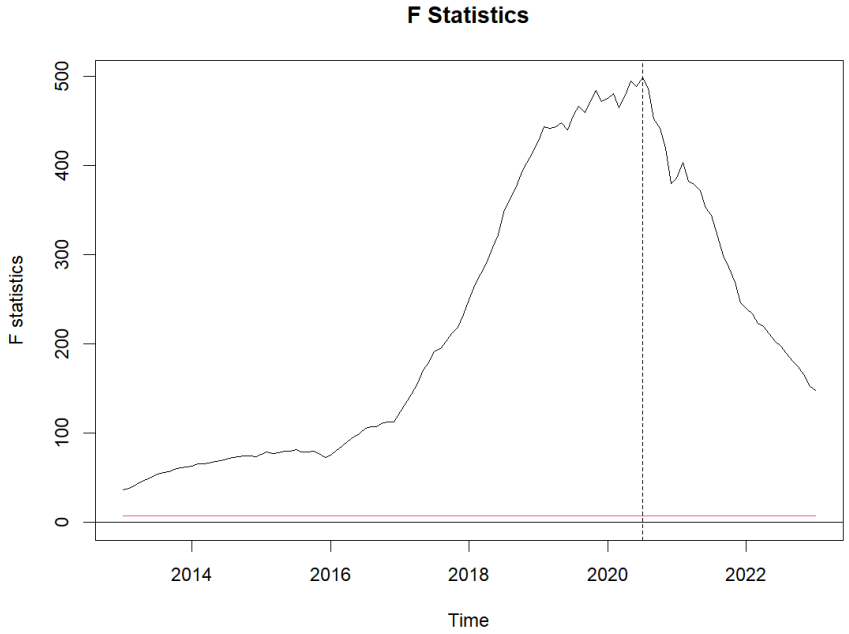
```
######################
# KPSS Unit Root Test #
######################

Test is of type: tau with 4 lags.

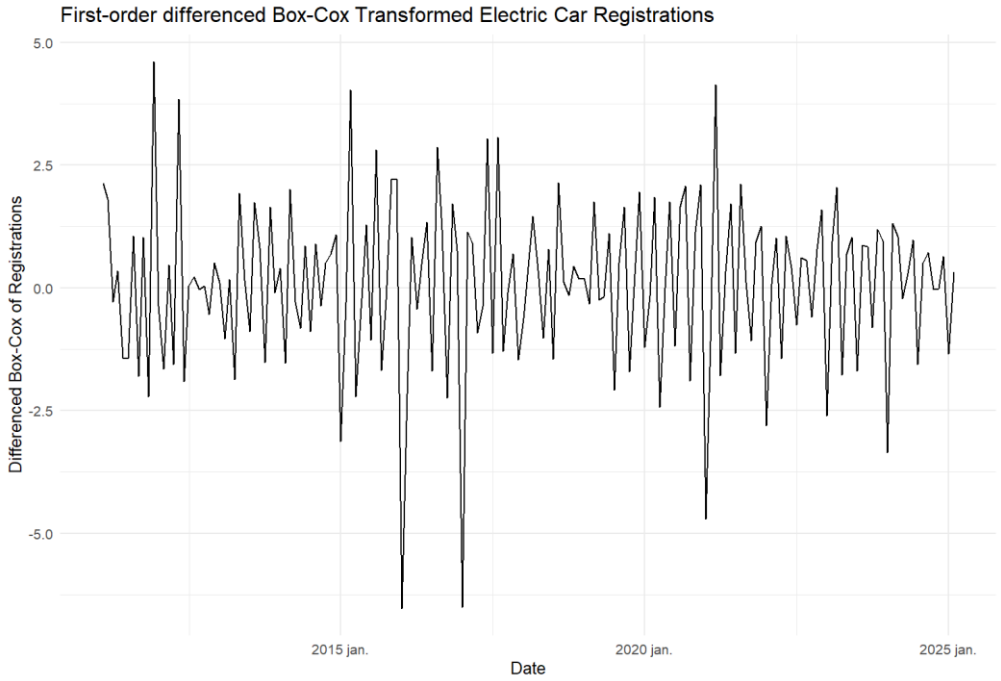
Value of test-statistic is: 0.5606

Critical value for a significance level of:
                10pct  5pct 2.5pct  1pct
critical values 0.119 0.146  0.176 0.216
```

QLR test applied on the Box-Cox transformed time series



First-order differenced Box-Cox transformed whole series



Results of KPSS and ADF test applied on the differenced Box-Cox transformed whole series

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression trend


Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)

Residuals:
    Min      1Q  Median      3Q     Max
-5.4848 -0.5876  0.0643  0.8030  3.8628

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.006742   0.221195   0.030    0.976
z.lag.1     -1.959173   0.114867 -17.056  < 2e-16 ***
tt           0.001574   0.002264   0.695    0.488
z.diff.lag   0.435519   0.069619   6.256 3.35e-09 ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.41 on 163 degrees of freedom
Multiple R-squared:  0.7463,    Adjusted R-squared:  0.7417
F-statistic: 159.9 on 3 and 163 DF,  p-value: < 2.2e-16


Value of test-statistic is: -17.056 96.9967 145.478

Critical values for test statistics:
      1pct  5pct 10pct
tau3 -3.99 -3.43 -3.13
phi2  6.22  4.75  4.07
phi3  8.43  6.49  5.47
```

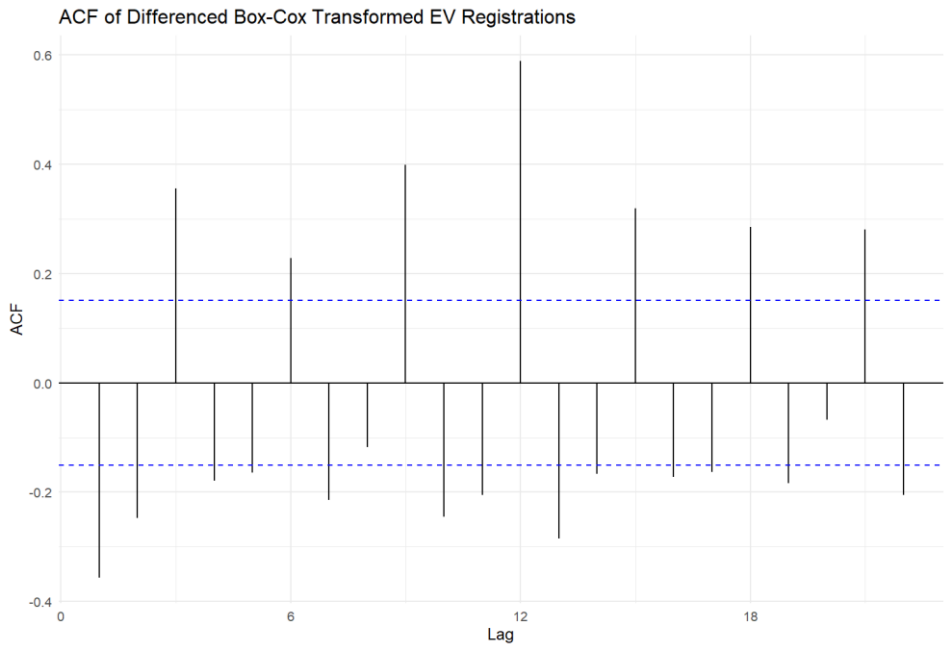
```
######################
# KPSS Unit Root Test #
######################

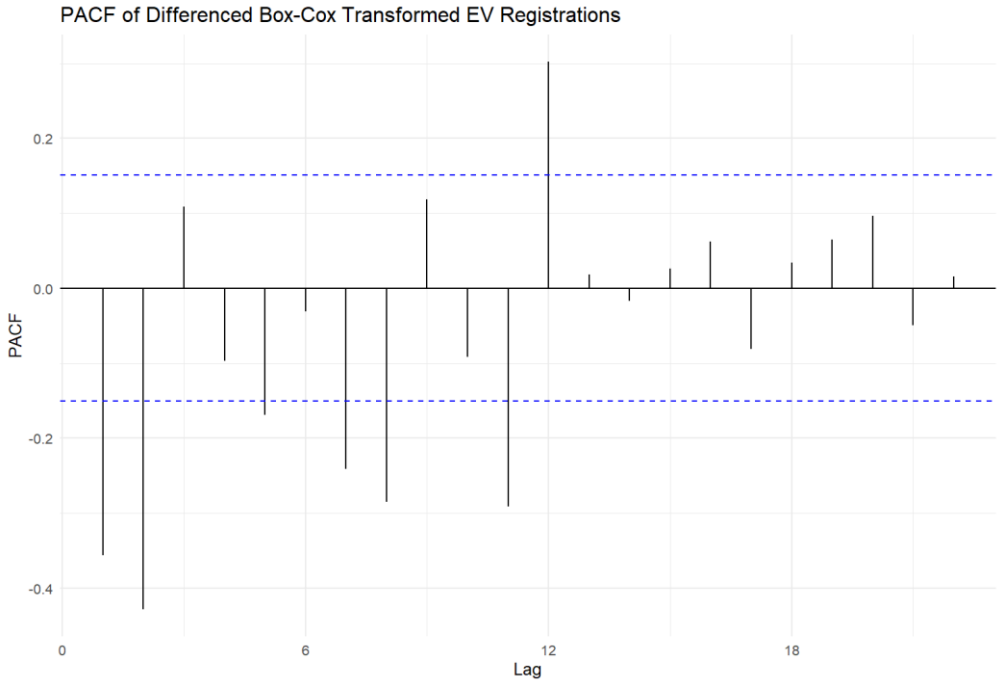
Test is of type: tau with 4 lags.

Value of test-statistic is: 0.022

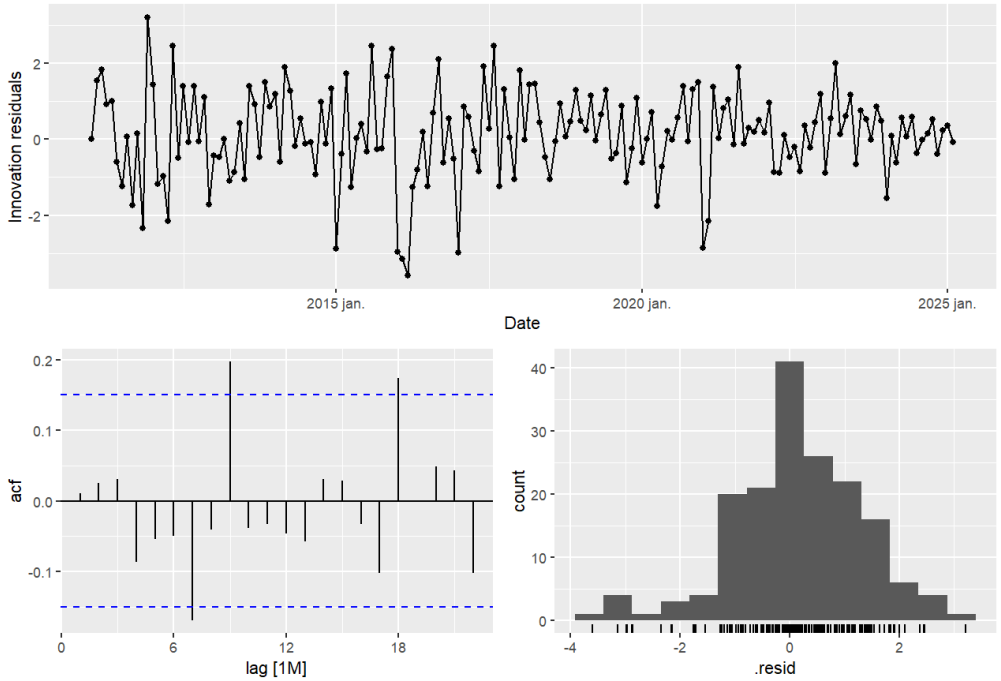
Critical value for a significance level of:
                10pct  5pct 2.5pct  1pct
critical values 0.119 0.146  0.176 0.216
```

ACF and PACF plot of the differenced Box-Cox transformed whole series

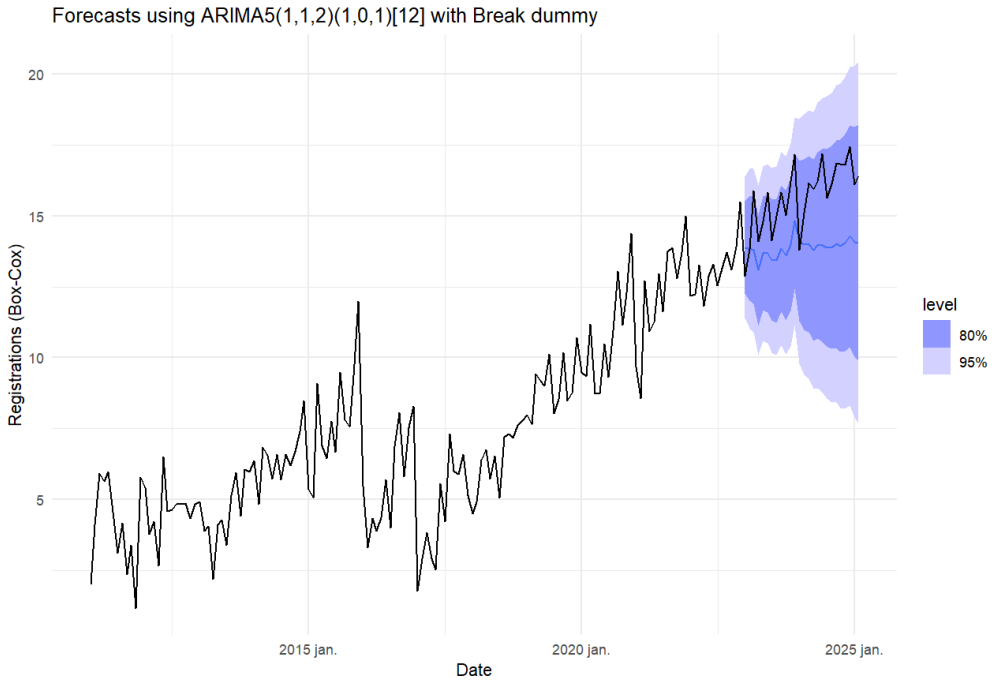


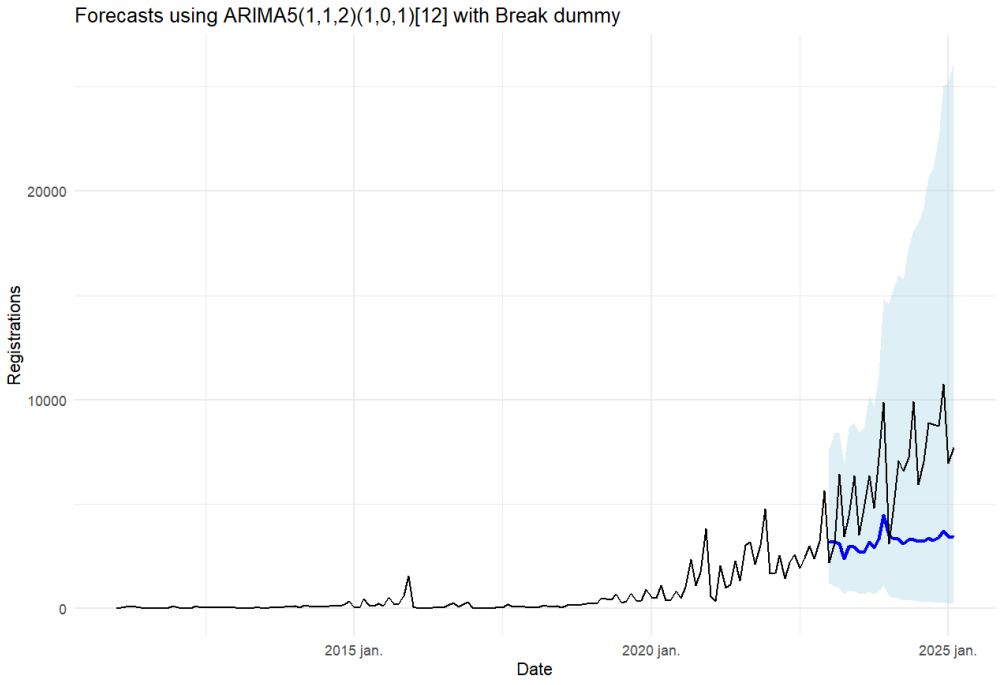


Residuals of ARIMA(1,1,2)(1,0,1)[12] with Break Dummy

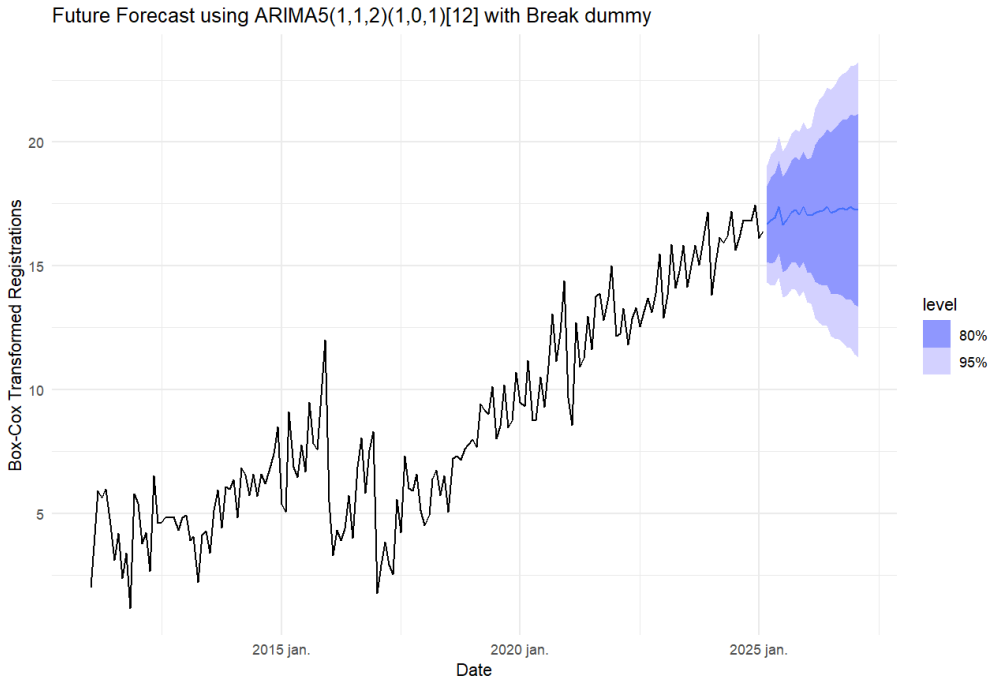


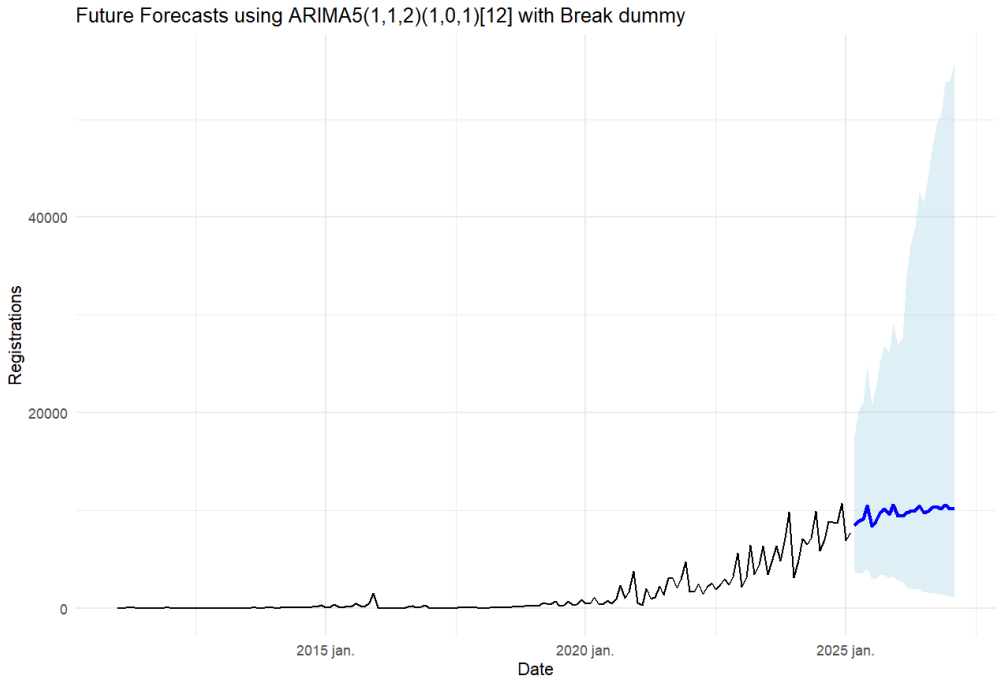
Forecast on training set using ARIMA(1,1,2)(1,0,1)[12] with Break Dummy



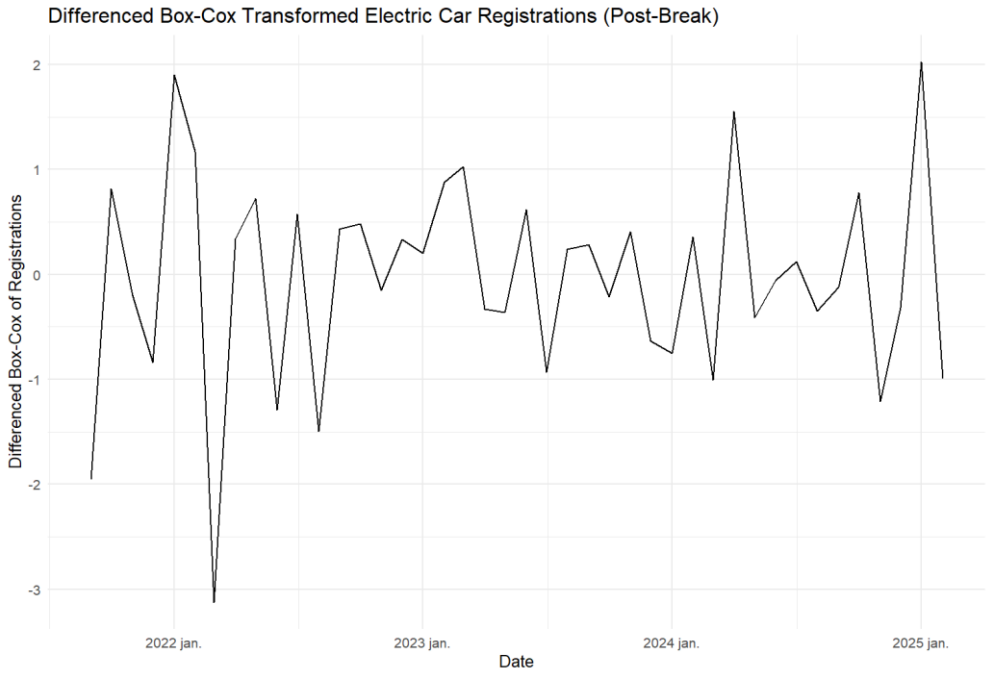


Future Forecast using ARIMA(1,1,2)(1,0,1)[12] with Break Dummy





Differenced Box-Cox transformed Post-Break Series



Results of KPSS and ADF test applied on the differenced Box-Cox transformed Post-Break series

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression trend


Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)

Residuals:
     Min       1Q   Median       3Q      Max
-1.72271 -0.53063 -0.04737  0.40839  1.83292

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.07561    0.28426  -0.266  0.79175
z.lag.1     -1.94684    0.24716  -7.877 2.41e-09 ***
tt           0.00271    0.01164   0.233  0.81721
z.diff.lag   0.43802    0.15100   2.901  0.00631 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8493 on 36 degrees of freedom
Multiple R-squared:  0.7377,    Adjusted R-squared:  0.7158
F-statistic: 33.74 on 3 and 36 DF,  p-value: 1.466e-10


Value of test-statistic is: -7.8769 20.684 31.024

Critical values for test statistics:
      1pct  5pct 10pct
tau3 -4.15 -3.50 -3.18
phi2  7.02  5.13  4.31
phi3  9.31  6.73  5.61
```

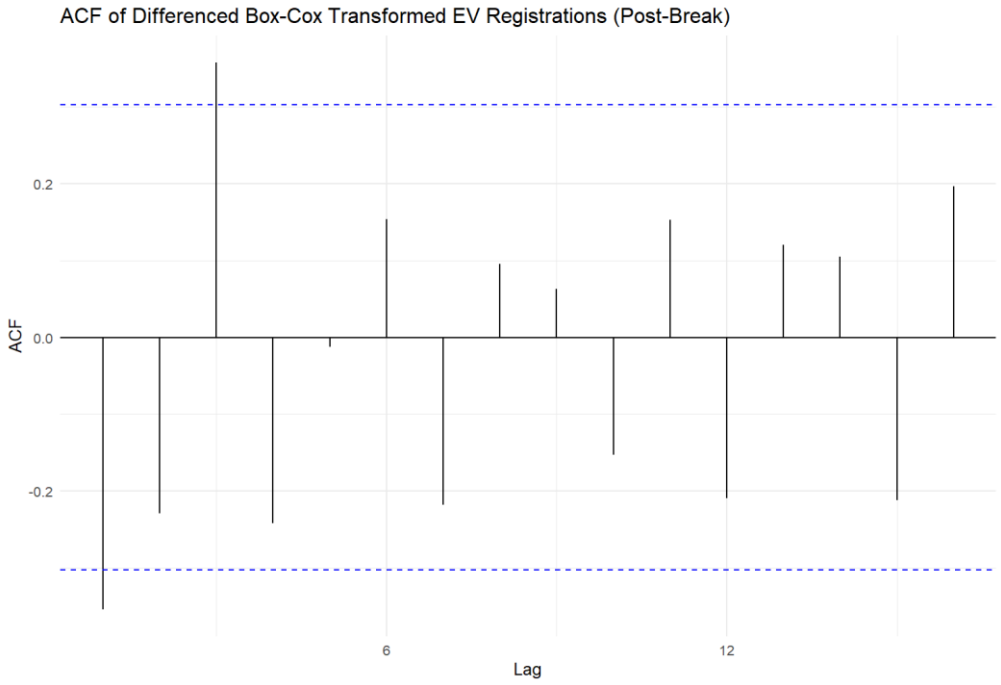
```
######################
# KPSS Unit Root Test #
######################

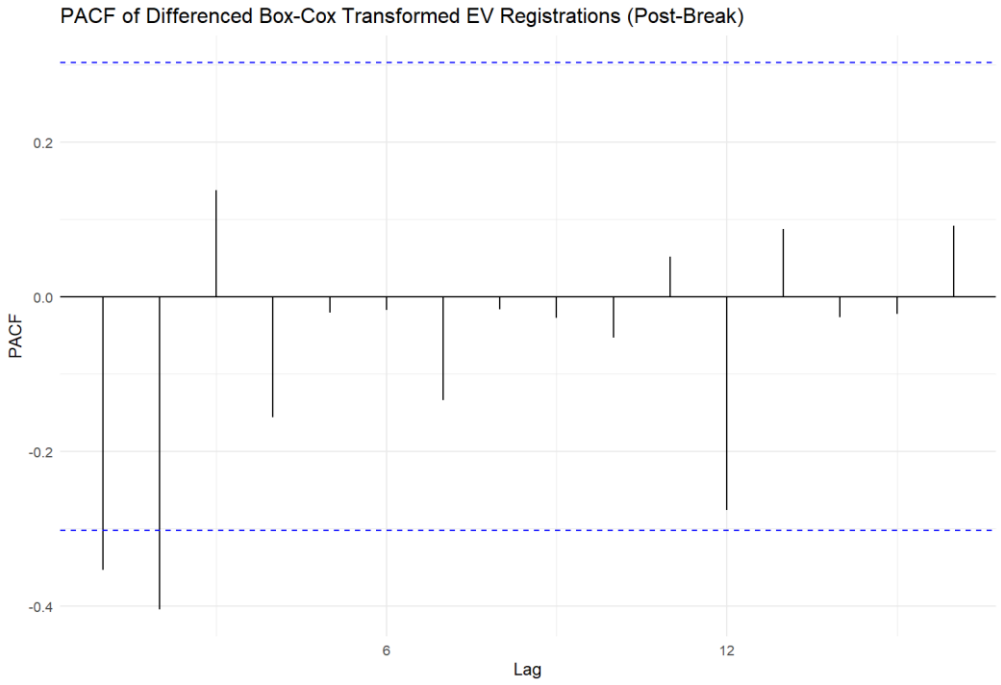
Test is of type: tau with 3 lags.

Value of test-statistic is: 0.0622

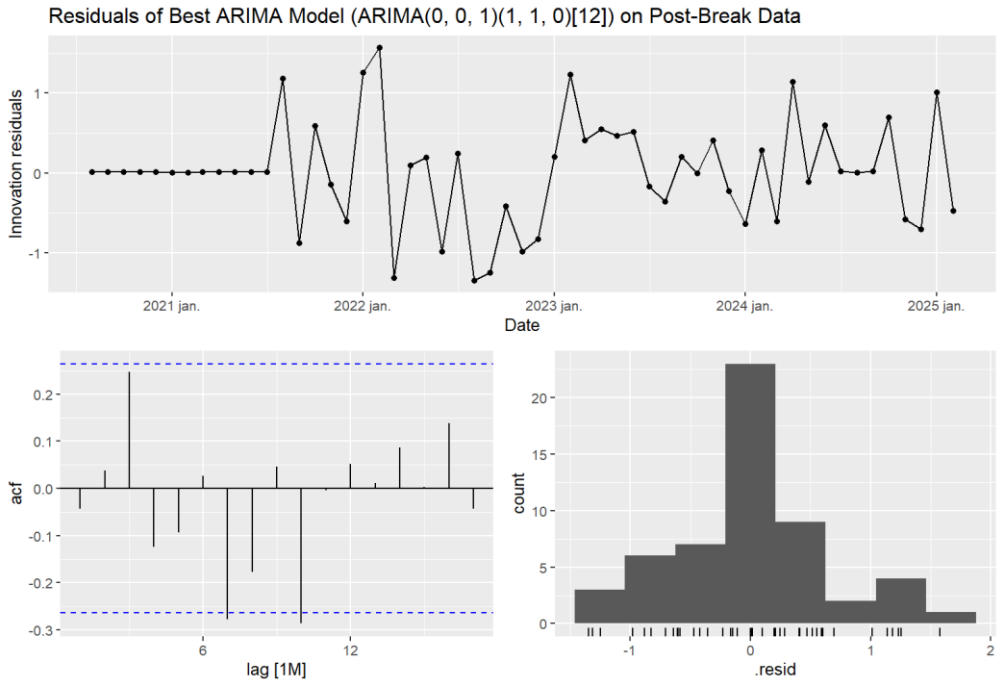
Critical value for a significance level of:
                10pct  5pct 2.5pct  1pct
critical values 0.119 0.146  0.176 0.216
```

ACF and PACF of the differenced Box-Cox transformed Post-Break Series

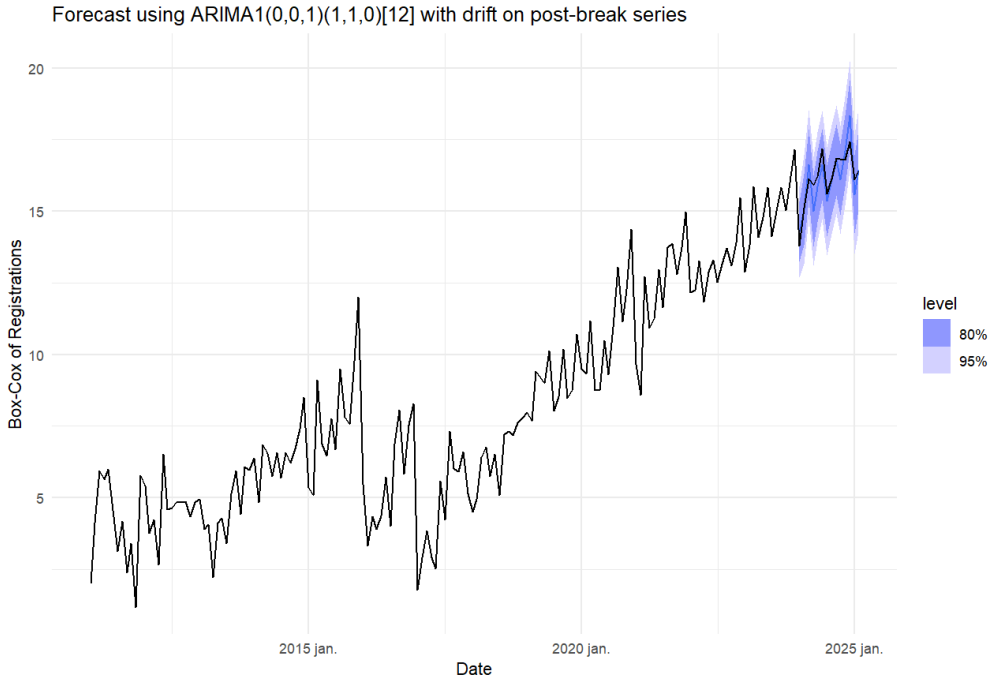


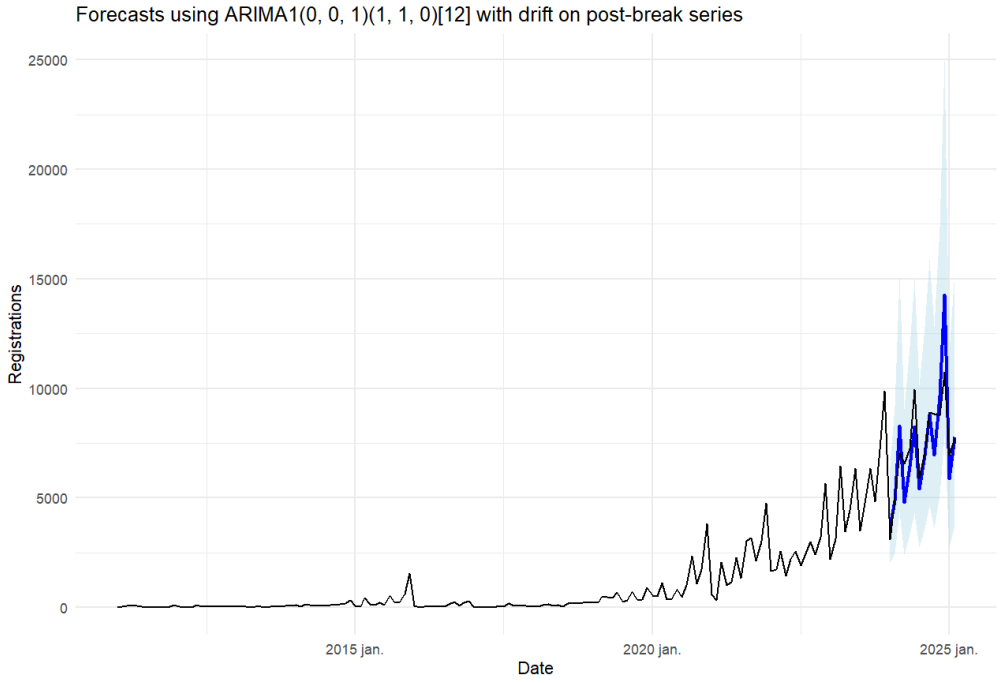


Residuals of ARIMA(0,0,1)(1,1,0)[12] with drift on Post-Break Series

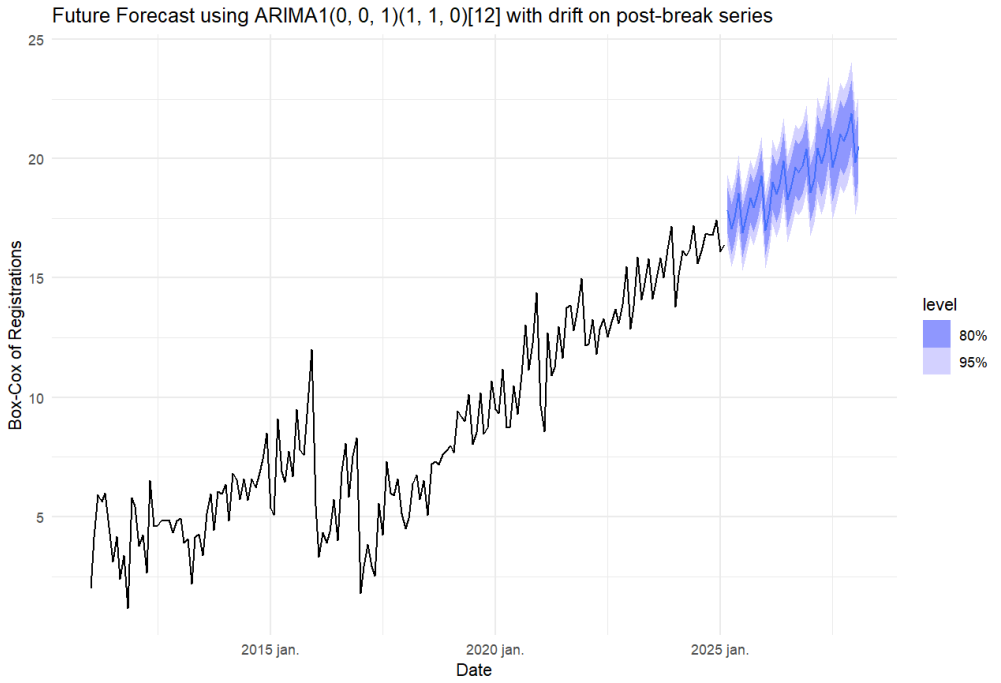


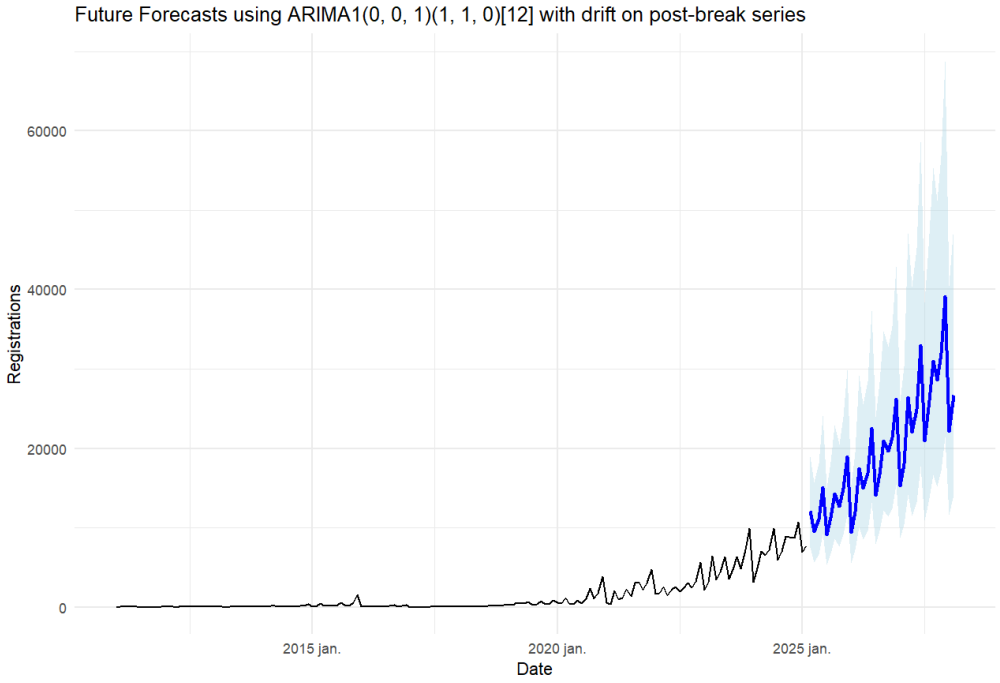
Forecasts on training set using ARIMA(0,0,1)(1,1,0)[12] on post-break series



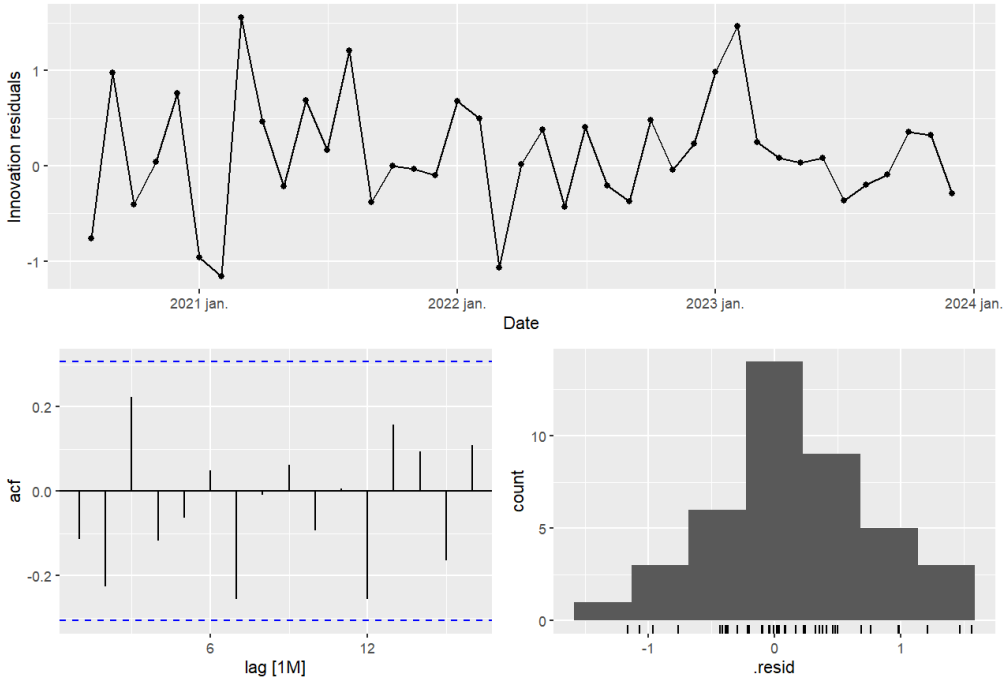


Future forecasts using ARIMA(0,0,1)(1,1,0)[12] with drift on post-break series

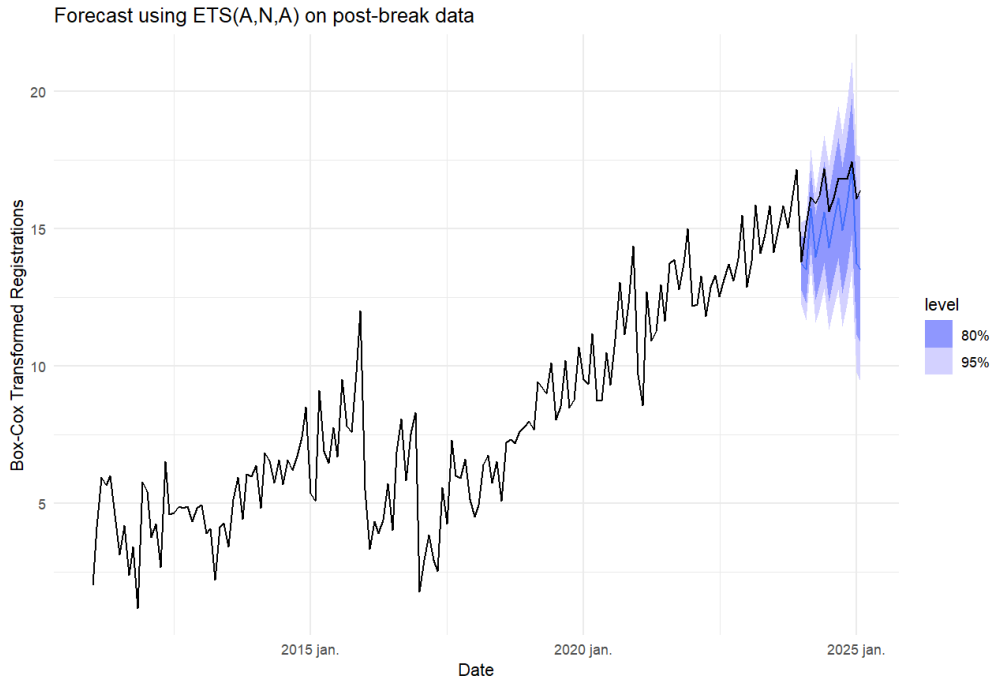


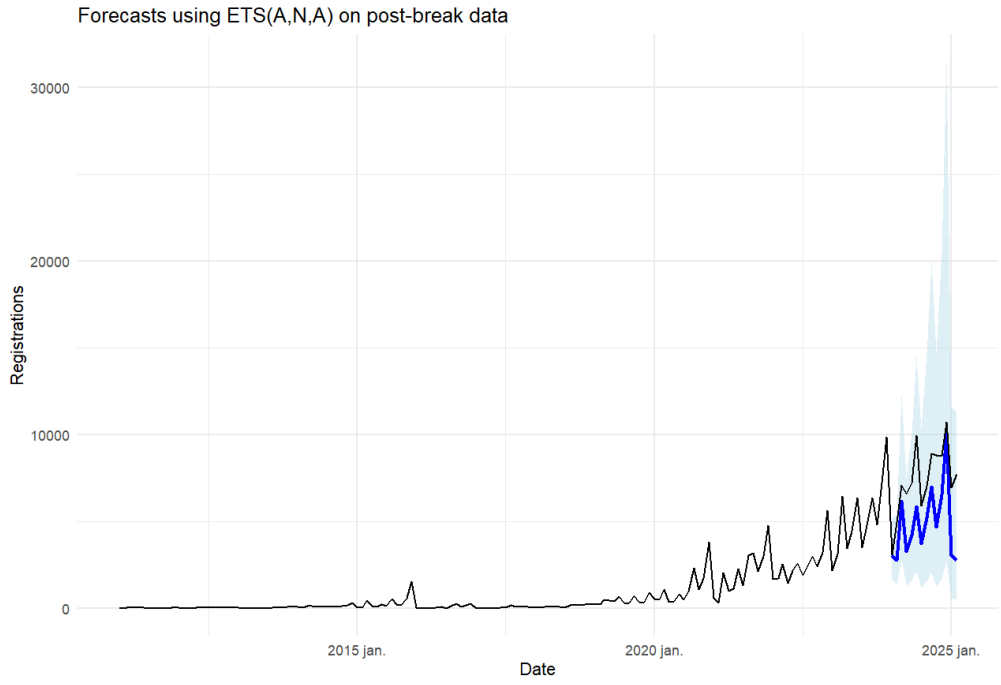


Residuals of ETS(A,N,A) model



Forecast on training set using ETS(A,N,A) on Post-Break data





Future forecast using ETS(A,N,A) on post-break data

